

ASSESSING VOLATILITY RISK AND SHOCK PERSISTENCE IN DOW JONES INDUSTRIAL AVERAGE STOCKS

HOMework TEAM 4

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Overview

The Commercial Banking Corporation (hereafter the “Bank”), acting by and through its Portfolio Pricing Department, has requested an assessment of stock volatility among the components of the Dow Jones Industrial Average (DJIA). Therefore, our team forecasted stock volatility among the top five most significant stock components of the DJIA using Autoregressive Conditional Heteroskedasticity (ARCH) and Generalized ARCH (GARCH) methodologies to support portfolio pricing and risk management decisions. Our analysis identified the following stocks as the most significant for volatility modeling:

- Goldman Sachs Group Incorporated (GS)
- Honeywell International Incorporated (HON)
- Procter & Gamble Company (PG)
- JPMorgan Chase & Company (JPM)
- Microsoft Corporation (MSFT)

Amongst these five stocks, GS stocks exhibited the highest sensitivity to market shocks, while HON demonstrated the longest recovery period following market shocks. Overall, our analysis indicated that DJIA represents a dependable investment area for the Bank’s services, as its most volatile assets provide a reliable combination of high risk and stability.

Methodology

The following section describes the structure and preparation of the data used to rank stock resilience and forecast stock volatility.

Data Used

Our team collected daily stock prices and returns from the 30 companies currently included in the DJIA from March 1st, 2011 to March 24th, 2024. The current DJIA portfolio consisted of major corporations such as Amazon, Walmart, Nike, and Cisco Systems.

Data Preprocessing

To prepare for our analysis, we calculated both simple returns and logarithmic returns for each stock using daily closing prices. We ensured that the data was in a tidy format, with prices and returns correctly aligned by date, to help facilitate accurate and efficient modeling of volatility patterns.

Analysis

This section details the selection of the most significant stocks in the DJIA, and describes the creation and evaluation of volatility models.

Stock Selection

Using ARCH/GARCH methodologies, we ranked the 30 stocks components of the DJIA based on their significance. The goal was to identify the stocks most appropriate for GARCH modeling and narrow the analysis to the five stocks with the strongest evidence of time-varying volatility.

We used the Lagrange Multiplier Lag 1 test to determine if a stock's squared residuals were significantly correlated with past values. Any stocks that produce significant results from this test indicate that GARCH modeling is appropriate, due to the presence of volatility clustering within that stock. Therefore, this test allowed us to identify the top five most significant stocks for ARCH/GARCH modeling. Table 1 lists these stocks in descending order of significance.

Table 1: Lagrange Multiplier Lag 1 Test Results

Ticker Symbol	Company	P-Value
GS	Goldman Sachs Group Incorporated	6.334×10^{-145}
HON	Honeywell International Incorporated	8.462×10^{-144}
PG	Procter & Gamble Company	3.553×10^{-141}
JPM	JPMorgan Chase & Company	3.774×10^{-136}
MSFT	Microsoft Corporation	6.719×10^{-116}

The stocks displayed in Table 1 exhibited the strongest evidence of time-varying volatility based on their respective p-values and were expected to provide the most meaningful insight. In other words, these five stocks contributed most strongly to the volatility behavior in the DJIA.

Model Creation & Selection

For each of the five selected stocks, multiple ARCH/GARCH models were developed to capture different characteristics of volatility behavior. These models vary in complexity, distributional assumptions, and their ability to account for asymmetry after market shocks. The following models were used for estimation and their characteristics are shown in Table 2.

Table 2: ARCH/GARCH Model Characteristics

Model	Distribution	Symmetry	Complexity	Key Characteristic
GARCH-normal	Normal distribution	Symmetric	Low	Underestimates extreme movements.
GARCH-t	T-distribution	Symmetric	Low	Accounts for heavy-tailed distributions.
EGARCH-t	T-distribution	Asymmetric	High	Captures asymmetric volatility patterns.
GJR-GARCH-t	T-distribution	Asymmetric	High	Allows for positive and negative shock responses.
APARCH-normal	Normal distribution	Asymmetric or Symmetric	High	Allows for varied volatility dynamics.

We decided to incorporate asymmetry through the EGARCH-t and GJR-GARCH-t models in Table 2.

Model performance for each stock was measured using the Bayesian Information Criterion (BIC). The model with the lowest BIC value was selected as the best model for volatility forecasting. Across all five stocks, the GJR-GARCH model had the lowest BIC values, indicating that the model did well at fitting the asymmetric volatility at the expense of model parsimony.

Results & Recommendations

This section describes the volatility forecasts and stock resilience amongst the five stocks from above.

To evaluate risk, we generated a 30-day volatility forecast for each stock using the GJR-GARCH-t model, which captures asymmetric responses to market shocks. The t-distribution was used to accommodate the heavy distribution tails of each stock. Figure 1 shows the 30-day forecast for all five stocks.

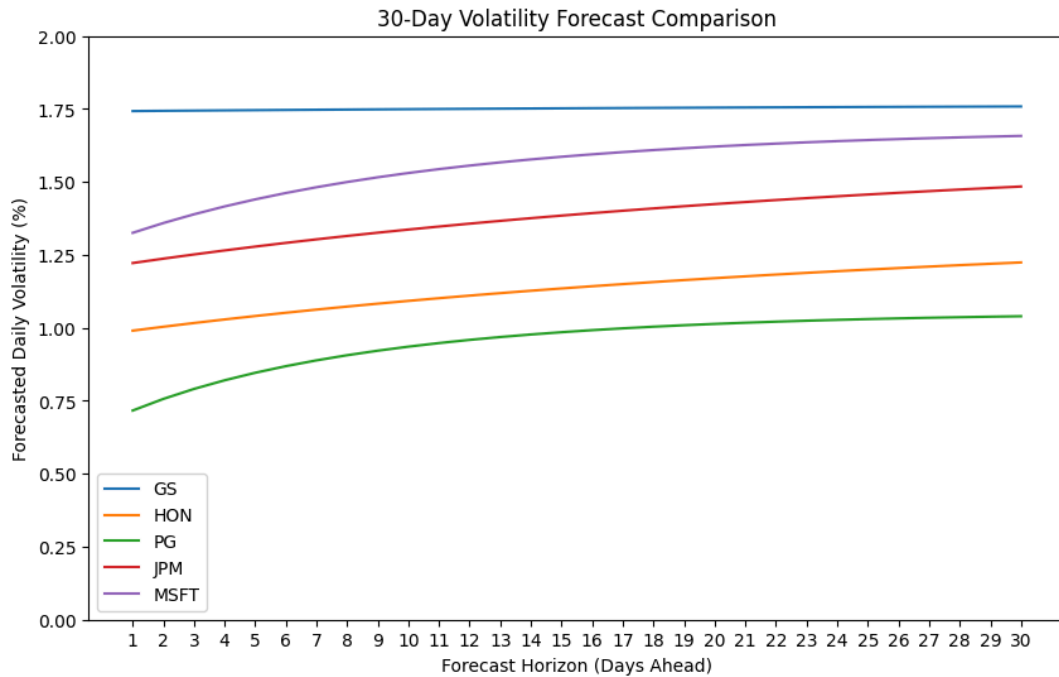


Figure 1: 30-Day Volatility Forecast Comparison

As seen in Figure 1, volatility increases over the projected 30 days with no sharp deviations.

Stock Ranking

From each of our models, we are able to pull alpha and beta parameters to get a better understanding of the volatility of each of these five stocks.

The alpha parameter for each stock's respective model is listed in Table 3 in descending order.

Table 3: Stock Sensitivity to Shocks

Ticker Symbol	Company	Alpha
GS	Goldman Sachs Group Incorporated	0.0394
MSFT	Microsoft Corporation	0.0273
PG	Procter & Gamble Company	0.0225
JPM	JPMorgan Chase & Company	0.0202
HON	Honeywell International Incorporated	0.0134

From Table 3, we saw that GS had the largest alpha parameter, indicating it is the most sensitive to market events. HON had the smallest alpha parameter, indicating that it was the least sensitive to market events.

In addition to pulling the alpha parameter from each stock's respective model, we also evaluated the beta parameter, which is displayed in descending order in Table 4.

Table 4: Stock Recovery Ranking

Ticker Symbol	Company	Beta
HON	Honeywell International Incorporated	0.9055
GS	Goldman Sachs Group Incorporated	0.8881
JPM	JPMorgan Chase & Company	0.8866
MSFT	Microsoft Corporation	0.8748
PG	Procter & Gamble Company	0.8740

Table 4 shows that HON had the largest beta parameter, which means that volatility for that stock takes the longest to decay following a market event when compared to the other four analyzed stocks. In contrast, PG had the smallest beta parameter, which means that it recovers most quickly.

Recommendations

As indicated by the alpha and beta parameters estimated from our GARCH models, the DJIA stocks exhibit a wide range of risk behaviors and volatility characteristics. Some stocks, such as GS, demonstrate high sensitivity to market shocks with a long recovery time, indicating that managing exposure to this asset may require repeated hedging strategies. Other stocks, such as PG, show lower sensitivity and faster recovery, implying a lower risk and more predictable volatility. Given this diversity, the DJIA represents a dependable investment area for the bank, offering opportunities for both high-return, higher-risk positions and more stable, lower-risk growth allocations.

Conclusion

Overall, we'd recommend DJIA for banking services, as these five stocks seem to be predictably volatile together. Though this group of stocks include those that experience amplified shock sensitivity and overall higher risk, such as GS, it also includes low-sensitivity stocks like HON and JPM. These latter stocks would be good candidates for long-term investments. Though these five stocks are only a small portion of the total 30 in the Dow, they capture a large share of DJIA conditional variance behavior. Whether its investments, benchmarking, or future modeling, our analysis shows that DJIA would be reliable enough for the Bank's services.